

Eigenvalues of tetrahedral sums of Hermitian matrices

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Abstract

This is the abstract.

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1 Introduction

Notational conventions: Throughout this paper, $d \in \mathbb{N}_{>0}$ will denote the size of the matrices under consideration. We also exclusively work with matrices with coefficients in the field of complex numbers $\mathbb{C}$, so for notational brevity we write $\mathrm{GL}(d)$ as shorthand for $\mathrm{GL}(d, \mathbb{C})$. Weights for $\mathrm{GL}(d)$ will be denoted using Greek letters and will be identified with d -tuples of integers, e.g. $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{Z}^d$. The weight lattice for $\mathrm{GL}(d)$ is denoted by $\Lambda^{\mathbb{Z}}(d) \simeq \mathbb{Z}^d$ in the usual way; a weight $\alpha \in \Lambda^{\mathbb{Z}}(d)$ is dominant if $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_d$ and the set of all dominant weights for $\mathrm{GL}(d)$ is denoted by $\Lambda_{\geq}^{\mathbb{Z}}(d)$.

The set of all $d \times d$ complex Hermitian matrices will be denoted by $\mathcal{H}(d)$ while the set of all possible eigenvalues of a $d \times d$ complex Hermitian matrix is denoted by $\Lambda_{\geq}^{\mathbb{R}}(d) \subset \mathbb{R}^d$. The set of $d \times d$ Hermitian matrices X with eigenvalues $\mathrm{eig}(X) = (x_1, \dots, x_d) = \Lambda_{\geq}^{\mathbb{R}}(d)$ will be denoted by $\mathcal{O}_x$.

2 Generalizations of Horn's problem

This section will include a formal statement of the problem under investigation, its relation to Horn's original problem and other generalizations of Horn's problem. This section will hopefully briefly describe what is known about the eigenvalues of triangular sums of matrices. Regarding tetrahedral spectra, we will describe many of the symmetries of the problem statement and various “gauge” choices that one could make.

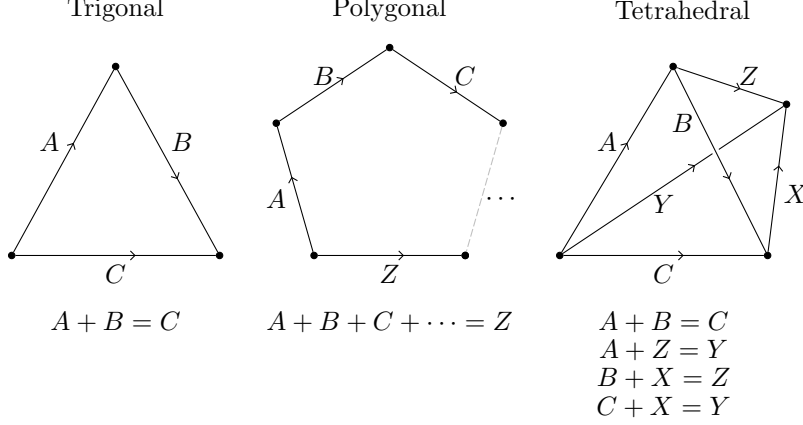


Figure 1: Examples of graphical Horn problems.

Additional generalizations of the problem statement, like to more matrices, linear combinations, etc. will be mentioned here. It will be important to mention the manner in which the analogue of the “Thompson-Klyachko” theorem does not hold, i.e. singular values of tetrahedral products are not the exponentials of eigenvalues of tetrahedral sums in general.

2.1 Graphical Horn problems

Definition 1. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite directed graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$ together with source and target maps $s, t : \mathcal{E} \rightarrow \mathcal{V}$. Let d be a positive integer. A *spectral map* (for $\mathcal{G}$ of dimension d) is a function $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ assigning to each edge $e \in \mathcal{E}$ in the directed graph a d -tuple of real numbers $\lambda(e) = (\lambda_1(e), \dots, \lambda_d(e))$. The map $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is said to be *realizable (by Hermitian matrices)* if there exists a map $H : \mathcal{E} \rightarrow \mathcal{H}(d)$ assigning to each edge $e \in \mathcal{E}$ a $d \times d$ Hermitian matrix $H(e)$ such that

- for each edge $e \in \mathcal{E}$, $H(e)$ has eigenvalues $\lambda(e)$,

$$\text{eig}(H(e)) = \lambda(e), \tag{1}$$

- for each cycle $\ell \subseteq \mathcal{E}$ with orientation map $\gamma_{\ell} : \ell \rightarrow \{\pm 1\}$,

$$\sum_{e \in \ell} \gamma_{\ell}(e) H(e) = 0. \tag{2}$$

Alternatively, one can equivalently encode the constraints imposed by Eq. (2) in terms of a $\{-1, 0, +1\}$ -valued $L \times |E|$ matrix (where L is the number of (generating) cycles in $\mathcal{G}$), denoted by $\Gamma(\mathcal{G})$, where

$$\Gamma(\mathcal{G})_{\ell, e} = \begin{cases} 0 & e \notin \ell, \\ \gamma_{\ell}(e) & e \in \ell. \end{cases} \tag{3}$$

In terms of $\Gamma(\mathcal{G})$, Eq. (2) reads for all cycles ℓ :

$$\sum_{e \in \mathcal{E}} \Gamma(\mathcal{G})_{\ell, e} H(e) = 0 \tag{4}$$

Problem 1 (Graphical Horn problem). Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite directed graph, let d be a positive integer. The *graphical Horn problem* is to characterize the set of realizable spectral maps $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ for $\mathcal{G}$ of dimension d .

There might be an important connection to the their of quivers and representations of quivers. In particular, the notion of a morphism between a representation of a quiver in terms of operators on vertices forming computing squares models precisely the kinds of continuous symmetries which may be present in a given graphical Horn problem. Indeed, the spectral data were are ultimately interested in is equivalent to the orbit under such “local” isomorphisms (i.e. conjugation by isomorphisms).

TC: It turns out that the only relevant data for statement of a graphical Horn problem is the matrix $\Gamma(\mathcal{G})$; perhaps its best to drop reference to $\mathcal{G}$ altogether and just have the problem statement depend on a matrix Γ ? Moreover, one could in principle take the matrix to be $\mathbb{R}$ -valued instead of $\{-1, 0, +1\}$ -valued, although that would make the connection to the representation theory even weaker. Generalizing to $\mathbb{Z}$ -valued Γ is also essentially the same I think...

There are a few operations which preserve the set of realizable spectral maps.

Lemma 1. *Consider a graphical Horn problem of dimension d .*

- *The zero spectral map $0_d : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ assigning to $e \in \mathcal{E}$, the spectrum $0_d(e) = (0, \dots, 0)$ is realizable.*
- *If $\alpha : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(a)$ and $\beta : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(b)$ are realizable, then their direct sum $\alpha \oplus \beta : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(a+b)$ and tensor product $\alpha \otimes \beta : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(ab)$ (defined pointwise in the obvious manner), are both realizable.*
- *If $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable and $p \in \mathbb{R}_{>0}$ a positive real number, then $p\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$, defined by $(p\lambda)(e) = (p\lambda_1(e), \dots, p\lambda_d(e))$ is also a realizable.*
- *If $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable, then $\lambda^* : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$, defined by a pointwise application of the negation-reversal involution, is also realizable.*

Example 1 (The trigonal Horn problem). Perhaps the smallest non-trivial example of a graphical Horn problem is the one for which Horn originally conjectured a solution [Hor62], namely, what are the possible eigenvalues, (α, β, γ) , of a triple of $d \times d$ Hermitian matrices (A, B, C) satisfying the condition $A + B = C$? From the perspective of Problem 1, we refer to this question as the *trigonal Horn problem*.

Example 2 (Polygonal Horn problems). If the directed graph $\mathcal{G}$ consists of a single closed cycle, then the corresponding Horn problem will be referred to as a *polygonal Horn problem*; the polygonal Horn problem associated to an *oriented* cycle $e_1 \rightarrow e_2 \rightarrow \dots \rightarrow e_r \rightarrow e_1$ consisting of r edges asks what are the possible eigenvalues $(\lambda_1, \dots, \lambda_r)$ of r Hermitian matrices $(H_1, \dots, H_r)$ with zero-sum $H_1 + \dots + H_r = 0$. Evidently, there is a bijection between the set of realizable spectra (α, β, γ) and the set of realizable spectra $(\lambda_1, \lambda_2, \lambda_3)$ for the polygonal Horn problem where $r = 3$: namely

$$(\lambda_1, \lambda_2, \lambda_3) = (\alpha, \beta, \gamma^*) \quad (5)$$

where $\gamma \mapsto \gamma^*$ is the involution on $\Lambda_{\geq}^{\mathbb{R}}(d)$ defined by negation of entries and then reversing to preserve the ordering:

$$\gamma^* = (-\gamma_d, \dots, -\gamma_1). \quad (6)$$

Example 3 (Tetrahedral Horn problem). In this paper, a considerable focus will be placed on the *tetrahedral Horn problem* which is the graphical Horn problem associated with the directed graph above forming a tetrahedron. Specifically, a sextuple of candidate spectra,

$$(\alpha, \beta, \gamma, \mu, \nu, \lambda) \in (\Lambda_{\geq}^{\mathbb{R}}(d))^{\times 6} \quad (7)$$

are said to be *tetrahedral* if they are the respective eigenvalues of Hermitian matrices

$$(A, B, C, M, N, L) \in \mathcal{H}(d)^{\times 6} \quad (8)$$

satisfying the following four conditions:

$$A + B = M, \quad M + C = L, \quad B + C = N, \quad A + N = L. \quad (9)$$

Example 4 (One-dimensional graphical Horn problems). An important special case of graphical Horn problem for a digraph $\mathcal{G}$ is the case where $d = 1$. If $d = 1$, then vector space of 1×1 Hermitian matrices becomes isomorphic to the real numbers, $\mathcal{H}(1) \simeq \mathbb{R}$, and similarly the space of candidate eigenvalues becomes $\Lambda_{\geq}^{\mathbb{R}}(1) = \mathbb{R}$. Moreover, the spectral map $\lambda : \mathcal{E} \rightarrow \mathbb{R}$ becomes an assignment of real numbers of the edges of $\mathcal{G}$, the map $H : \mathcal{E} \rightarrow \mathbb{R}$ must necessarily equal spectral map λ by the spectral condition Eq. (1), and the cycle condition characterizing realizable spectral maps simplifies to the condition that λ , viewed as a vector $(\lambda(e))_{e \in \mathcal{E}} \in \mathbb{R}^{|\mathcal{E}|}$, must lie in the kernel of $\Gamma(\mathcal{G})$,

$$\lambda \in \ker(\Gamma(\mathcal{G})). \quad \text{\texttt{eq:ker_cond_d_1}} \quad (10)$$

In fact, this simple condition provides a complete solution to all graphical Horn problems where $d = 1$.

The solution to all one-dimensional graphical Horn problems provided by Eq. (10) also gives rise to a simple necessary condition for $d > 1$ graphical Horn problems.

Lemma 2. *Consider the graphical Horn problem of dimension d for the digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Then the spectral map $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is realizable only if*

$$\text{Tr} \circ \lambda \in \ker(\Gamma(\mathcal{G})), \quad (11)$$

where, by abuse of notation, $\text{Tr} : \Lambda_{\geq}^{\mathbb{R}}(d) \rightarrow \mathbb{R}$ denotes the sum

$$\text{Tr}((\lambda_1, \dots, \lambda_d)) := \lambda_1 + \dots + \lambda_d. \quad (12)$$

Proof. Suppose that $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a spectral map realized by the assignment $H : \mathcal{E} \rightarrow \mathcal{H}(d)$, meaning $\text{eig}(H(e)) = \lambda(e)$ for all edges $e \in \mathcal{E}$, and $\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} H(e) = 0$ for all cycles ℓ . Taking the trace of cycle condition Eq. (2) yields

$$0 = \sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \text{Tr}(H(e)) = \sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \text{Tr}(\lambda(e)). \quad (13)$$

In other words, $\text{Tr} \circ \lambda$ lies in the kernel of $\Gamma(\mathcal{G})$ which implies $\text{Tr} \circ \lambda$ is a realizable spectral map for the $d = 1$ case. \texttt{lem:ker_shift} $\square$

Lemma 3. *Consider the graphical Horn problem of dimension d for the digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Suppose $\mu : \mathcal{E} \rightarrow \mathbb{R}$ lies in the kernel of $\Gamma(\mathcal{G})$ and define $\mu_d : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ to be the spectral map defined by*

$$\mu_d(e) := (\mu(e), \dots, \mu(e)). \quad (14)$$

Then $\mu_d : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable spectral map. Moreover, $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is a realizable spectral map if and only if $\lambda + \mu_d$ is a realizable.

Proof. The spectral map μ_d is realized by the assignment $H_{\mu} : \mathcal{E} \rightarrow \mathcal{H}(d)$ defined by

$$H_{\mu}(e) := \mu(e)I_d \quad (15)$$

where I_d is the $d \times d$ identity matrix. This is because it was assumed that $\mu \in \ker(\Gamma(\mathcal{G}))$ and therefore

$$\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} H_{\mu}(e) = \left(\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \mu(e) \right) I_d = 0. \quad (16)$$

Now suppose $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is realized by the assignment $H_{\lambda} : \mathcal{E} \rightarrow \mathcal{H}(d)$, then it follows that $H_{\lambda} + H_{\mu}$, defined by

$$(H_{\lambda} + H_{\mu})(e) = H_{\lambda}(e) + \mu(e)I_d \quad (17)$$

realizes the spectral map $\lambda + \mu_d$ because the spectrum of $H_{\lambda}(e) + \mu(e)I_d$ is simply $\lambda(e) + \mu_d(e)$ (the eigenvalues add because $[H_{\lambda}(e), I_d] = 0$). $\square$

A direct corollary of Lemma 3 is that one can always assume, without loss of generality, that the spectral map $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ is traceless in the sense that $\text{Tr} \circ \lambda = 0_d$.

Corollary 4. Let $\lambda : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ be a spectral map such that $\text{Tr} \circ \lambda \in \ker(\Gamma(\mathcal{G}))$ and let $\lambda^0 : \mathcal{E} \rightarrow \Lambda_{\geq}^{\mathbb{R}}(d)$ be the traceless spectral map defined by

$$\lambda^0(e) := (\lambda_1(e) - \text{Tr}(\lambda(e)), \dots, \lambda_d(e) - \text{Tr}(\lambda(e))). \quad (18)$$

Then λ is realizable if and only if λ^0 is realizable.

Proof. The spectral map λ^0 is obtained from λ by subtracting the spectral map $\text{Tr} \circ \lambda$. By assumption, $\text{Tr} \circ \lambda$ lies in the kernel of $\Gamma(\mathcal{G})$ by Lemma 2, and thus Lemma 3 (when $\mu \mapsto \pm \text{Tr} \circ \lambda$) implies λ^0 is realizable if and only if λ is realizable. $\square$

3 Recoupling theory

The purpose of this section is to widen the bridge between representation theory and Horn-type problems. We will begin with a treatment of the connection between triangular spectra and the Littlewood-Richardson coefficients and discuss its connection to symplectic geometry and moment maps. Then we will articulate the manner in which tetrahedral spectra are related to the norm of the recoupling operator for $\text{SU}(d)$ and discuss the manner in which convexity results for moment maps cannot be applied and thus one must not expect the space of tetrahedral spectra to be convex (indeed it is not).

3.1 Triangular spectra & Littlewood-Richardson coefficients

[Lid82]

Definition 2. Let $\alpha, \beta, \gamma \in \Lambda_{\geq}^{\mathbb{Z}}(d)$ be a triple of dominant weights for $\text{GL}(d)$ and let $V_{\alpha}, V_{\beta}, V_{\gamma}$ be the triple of irreducible representations of $\text{GL}(d)$ with corresponding highest weights. Let $C_{\alpha, \beta}^{\gamma}$ denote the multiplicity space of $\text{GL}(d)$ -intertwiners from V_{γ} to $V_{\alpha} \otimes V_{\beta}$:

$$C_{\alpha, \beta}^{\gamma} := \text{Hom}_{\text{GL}(d)}(V_{\gamma}, V_{\alpha} \otimes V_{\beta}). \quad (19)$$

The *Littlewood-Richardson coefficient* is defined as

$$c_{\alpha, \beta}^{\gamma} := \dim(C_{\alpha, \beta}^{\gamma}). \quad (20)$$

3.2 Tetrahedral spectra & recoupling operators

3.3 An entropy inequality for tetrahedral spectra

4 The spectral quantum marginal problem

The purpose of this section will be to re-articulate the spectral embedding defined in [CŞW18] and state the sense in which the tetrahedral spectra problem is an instance of the quantum marginal problem. From here we can discuss the extent to which known inequalities coming from quantum information theory yield constraints necessarily satisfied by tetrahedral spectra.

4.1 A spectral embedding

`defn:spec_embed`

Definition 3. Let $(H_1, \dots, H_k) \in \mathcal{H}(d)^{\times k}$ be a k -tuple of $d \times d$ positive-semidefinite Hermitian matrices and let X be a subset of $\{1, \dots, k\}$. Let $\{|1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^d$ and $\{|0\rangle, |1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^{d+1}$. Define $H_X := \sum_{i \in X} H_i$ as the partial sum. Define a Hermitian operator, Φ_X , acting on $(\mathbb{C}^{d+1})^{\otimes k}$ with coefficients with respect to the computational bases $|i_1 \dots i_k\rangle$ be given by the

following formula:¹

$$\langle i_1 \cdots i_k | \Phi_X | j_1 \cdots j_k \rangle = \begin{cases} \langle i_p | H_p^{\frac{1}{2}} H_q^{\frac{1}{2}} | j_q \rangle & \exists! i_p, j_q > 0, p, q \in X \\ 0 & \text{otherwise.} \end{cases} \quad (21)$$

Note that, up to shuffling the $k+1$ copies of $\mathbb{C}^{d+1}$ in $(\mathbb{C}^{d+1})^{\otimes k}$, Φ_X is of the form $\tilde{\Phi}_X \otimes Z_{[k] \setminus X}$ where

$$Z_X = |0 \cdots 0\rangle \langle 0 \cdots 0|. \quad (22)$$

and where $\tilde{\Phi}_X$ acts on $(\mathbb{C}^{d+1})^{\otimes |X|}$. Also define the operator Γ_X which is $\tilde{\Phi}_X$ shifted by Z_X

$$\Gamma_X := \tilde{\Phi}_X + (1 - \text{Tr}(H_X))Z_X, \quad (23)$$

and in particular $\Gamma_\emptyset := 1$.

Example 5. Given two Hermitian operators $(H_1, H_2) \in \mathcal{H}(d)^{\times 2}$, the matrix $\Gamma_{1,2}$ has the following coefficients for all $i, j, k, l \in \{1, \dots, d\}$.

$$\langle 00 | \Gamma_{1,2} | 00 \rangle = 1 - \text{Tr}(H_{1,2}), \quad (24)$$

$$\langle i0 | \Gamma_{1,2} | j0 \rangle = \langle i | H_1 | j \rangle, \quad (25)$$

$$\langle 0i | \Gamma_{1,2} | 0j \rangle = \langle i | H_2 | j \rangle, \quad (26)$$

$$\langle i0 | \Gamma_{1,2} | 0j \rangle = \langle i | H_1^{\frac{1}{2}} H_2^{\frac{1}{2}} | j \rangle, \quad (27)$$

$$\langle 0i | \Gamma_{1,2} | j0 \rangle = \langle i | H_2^{\frac{1}{2}} H_1^{\frac{1}{2}} | j \rangle, \quad (28)$$

$$\langle ik | \Gamma_{1,2} | jl \rangle = 0. \quad (29)$$

lem:spec_embed

Lemma 5. Let everything be defined as in Definition 3. The follow statements hold for all subsets $X \subseteq [k]$.

(i) Up to padding by zeros, the eigenvalues of $\tilde{\Phi}_X$, Φ_X and H_X are all identical.

$$\lambda(\tilde{\Phi}_X) \simeq \lambda(\Phi_X) \simeq \lambda(H_X). \quad (30)$$

(ii) The eigenvalues of Γ_X are equal to the eigenvalues of $\tilde{\Phi}_X$ plus the additional value of $1 - \text{Tr}(H_X)$.

$$\lambda(\Gamma_X) \simeq \lambda(\tilde{\Phi}_X) \oplus \{1 - \text{Tr}(H_X)\}. \quad (31)$$

(iii) The partial trace of $\Gamma_{[k]}$ over $[k] \setminus X$ is equal to Γ_X :

$$\text{Tr}_{[k] \setminus X}(\Gamma_{[k]}) = \Gamma_X. \quad (32)$$

Proof. (i) From the definition of Φ_X , one observes that $\langle i_1 \cdots i_k | \Phi_X | i_1 \cdots i_k \rangle$ is only non-zero if there exists a unique non-zero $i_p \neq 0$ where $p \in X$ and thus

$$\text{Tr}(\Phi_X) = \sum_{p \in X} \sum_{j=1}^d \langle j | H_p | j \rangle = \sum_{p \in X} \text{Tr}(H_p) = \text{Tr}(H_X). \quad (33)$$

More generally, for any positive integer $\ell \in \mathbb{N}_{>0}$, one observes that $\text{Tr}(\Phi_X^\ell) = \text{Tr}(H_X^\ell)$. (**TODO: Prove this directly.**) As the eigenvalues of any matrix X are determined (up to padding by zeros) by the values of $\text{Tr}(X^\ell)$, we conclude that Φ_X and H_X have the same eigenvalues (up to padding by zeros) (**TODO: Is there a more direct way to prove this.**)

¹Specifically, $\langle i_1 \cdots i_k | \Phi_X | j_1 \cdots j_k \rangle$ is non-zero only if there exists a unique $p \in [k]$ such that $i_p > 0$ and a unique $q \in [k]$ such that $j_q > 0$ (with all other indices $i_{p'}$ and $j_{q'}$ for $p' \neq p$ and $q' \neq q$ being zero). Furthermore, if $p, q \in [k]$ are the unique such indices, then $\langle i_1 \cdots i_k | \Phi_X | j_1 \cdots j_k \rangle$ is zero if either $p \notin X$ or $q \notin X$; otherwise $p, q \in X$ and then $\langle i_1 \cdots i_k | \Phi_X | j_1 \cdots j_k \rangle$ takes the value $\langle i_p | H_p^{\frac{1}{2}} H_q^{\frac{1}{2}} | j_q \rangle$, the inner product between the i_p th column of $H_p^{\frac{1}{2}}$ and the j_q th column of $H_q^{\frac{1}{2}}$.

(ii) Note that $\tilde{\Phi}_X$ and its eigenvectors are supported in the subspace of $(\mathbb{C}^{d+1})^{\otimes |X|}$ spanned by $|i_1 \cdots i_{|X|}\rangle$ for which exactly one i_p is non-zero. Therefore $|0 \cdots 0\rangle$ is orthogonal to all eigenvectors of $\tilde{\Phi}_X$ and thus $|0 \cdots 0\rangle$ is an eigenvector of Γ_X with eigenvalue $1 - \text{Tr}(H_X)$.

(iii) The partial trace of $\Phi_{[k]}$ over $[k] \setminus X$, when performed in the computational basis $|i_1 \cdots i_k\rangle$, can be divided into three portions depending on the number of non-zero entries in $(i_p \mid p \in [k] \setminus X)$; first, is all indices i_p are zero, second, where exactly one i_p is non-zero, and third where more than one index i_p is non-zero. The respective contributions are as follows:

$$\text{Tr}_{[k] \setminus X}(\Phi_{[k]}) = \tilde{\Phi}_X + \text{Tr}(H_{[k] \setminus X})Z_X + 0 \quad (34)$$

Recalling the definition of $\Gamma_{[k]}$,

$$\Gamma_{[k]} := \Phi_{[k]} + (1 - \text{Tr}(H_{[k]}))Z_{[k]} \quad (35)$$

One concludes

$$\begin{aligned} \text{Tr}_{[k] \setminus X}(\Gamma_{[k]}) &= \text{Tr}_{[k] \setminus X}(\Phi_{[k]} + (1 - \text{Tr}(H_{[k]}))Z_{[k]}), \\ &= \tilde{\Phi}_X + \text{Tr}(H_{[k] \setminus X})Z_X + (1 - \text{Tr}(H_{[k]}))Z_X, \\ &= \tilde{\Phi}_X + (1 - \text{Tr}(H_X))Z_X, \\ &= \Gamma_X. \end{aligned} \quad (36)$$

□

Corollary 6. *By Lemma 5, one has $\text{Tr}(\Gamma_X) = 1$. If the eigenvalues of H_X satisfy*

$$\text{Tr}(H_X) \leq 1, \quad (37)$$

*then Γ_X is positive semidefinite, and thus Γ_X is a density matrix. In such a case, assuming $H_X \neq 0$ so that $\text{Tr}(H_X) \in (0, 1]$, the von Neumann entropy of Γ_X is related to the von Neumann entropy of $H_X / \text{Tr}(H_X)$ by the equation*²

$$S(\Gamma_X) = h(\text{Tr}(H_X)) + \text{Tr}(H_X)S\left(\frac{H_X}{\text{Tr}(H_X)}\right). \quad (38)$$

Moreover, if the eigenvalues of H_X satisfy the stronger condition

$$\lambda_1(H_X) + \text{Tr}(H_X) \leq 1, \quad (39)$$

then the maximum eigenvalue of Γ_X is $1 - \text{Tr}(H_X)$.

Proof. The only non-trivial claim involves the relation between the von Neumann entropies of Γ_X and $H_X / \text{Tr}(H_X)$, respectively given by

$$S(\Gamma_X) = -(1 - \text{Tr}(H_X))\log(1 - \text{Tr}(H_X)) - \sum_{i=1}^d \lambda_i(H_X) \log \lambda_i(H_X), \quad (40)$$

$$S\left(\frac{H_X}{\text{Tr}(H_X)}\right) = -\frac{1}{\text{Tr}(H_X)} \sum_{i=1}^d \lambda_i(H_X) \log \lambda_i(H_X) + \log(\text{Tr}(H_X)). \quad (41)$$

Multiplying the von Neumann entropy of $H_X / \text{Tr}(H_X)$ by $\text{Tr}(H_X)$ yields

$$\text{Tr}(H_X)S\left(\frac{H_X}{\text{Tr}(H_X)}\right) = S(\Gamma_X) + \text{Tr}(H_X)\log(\text{Tr}(H_X)) + (1 - \text{Tr}(H_X))\log(1 - \text{Tr}(H_X)), \quad (42)$$

which is the claimed result. □

5 Discussion

²Here $h(p) = -p \log p - (1 - p) \log(1 - p)$ denotes the binary entropy of $p \in [0, 1]$.

Reference notes

- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection to hyperbolic geometry quite explicitly.
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- An excellent expository overview of the big picture by Knutson [Knu00]
- The masters essay/thesis of Backens (TODO: check if the pdf Matthias sent to me is the final version) [Bac10]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]
- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- Klyachko [Kly00]
- Christandl [Chr08]
- Christandl, Şahinoğlu, and Walter [CSW18]
- Kirillov and Reshetikhin [KR90]
- Usage of Wigner’s $6j$ symbol to compute asymptotic error rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]

- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Boalch [Boa07]
- Taylor and Woodward [TW06]
- Henriques and Kamnitzer [HK05]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Potential application to inverse kinematics of 3-link arms in three dimensions? (Thanks Harold!)
- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko

- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).
- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Horn's original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn's inequalities in characterizing the existence of invariants vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]

Outline Draft May 30 2024

Below is the proposed outline of the paper as of May 30, 2024. Our proposed journal is *Communications in Mathematical Physics*. It will be ideal to illustrate with figures.

- introduction (include 2x2 case)
- general definition
- relationship to Horn's problem, (how does it generalize the problem)
- representation theoretic/quantization analogue (Ponzano-Regge and connection to state-sum models? etc.)
- comment about deviation from Thompson's conjecture (correspondence between eigenvalues of sums and logs of singularvalues of products)
- spectral embedding into the quantum marginal problem
- derivation of entropic inequalities (both from rep theory directly, and from embedding into QMP, things like concavity and data processing inequality for Renyi-entropies, etc), then prove they go beyond face constraints

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